IE 643: Deep Learning - Theory and Practice **July-Dec 2026**

Project Report: Cross-Template Key-Value Field Extraction from Scanned Forms

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Abstract

This report describes a document-understanding system that extracts key-value fields from scanned forms and is evaluated on form templates for which it has seen *zero* training examples. The task is posed without a schema: rather than being told which fields to look for, the model tags every word with a role (key, value, header or none) and then decides, for each value, which key on the page it answers. Both decisions are made by two small heads on a shared LayoutLMv3 backbone, trained jointly with a tagging loss and a per-value softmax linking loss. The central design choice is that the linking head sees geometry only as *relative* offsets in a scale-invariant frame, which we prove is invariant to translation and scaling of the page, and which an ablation shows is what makes the representation transfer. Training uses FUNSD, SROIE and WildReceipt under a template-level split with explicit leakage assertions; VRDU is held out entirely and never trained on. On unseen templates the system reaches 0.695 value recall against 0.918 on seen templates, and scores 0.593 micro F1 on VRDU's own benchmark-style field metric with no VRDU training data at all. A second split that holds out in-domain templates shows that entity, link and end-to-end pair F1 do *not* degrade on unseen templates — every gap interval contains zero — establishing that the headline gap is domain shift rather than template novelty. A failure analysis isolates the two structural assumptions that break: that a key sits beside its value, and that every value carries a printed label.

1 Introduction

Organisations receive the same information on endlessly different pieces of paper. A purchase order, an invoice, a government registration statement and a broadcast advertising contract all carry fields such as a date, a party's name and an amount, but each lays them out differently. The practical cost of this is that document-extraction systems are usually built *per template*: somebody collects examples of a form, annotates them, and trains or configures a model for that form. The moment a new layout arrives — a different vendor, a revised version of the same government form — the process repeats.

This project asks whether that per-template cost can be avoided. Concretely: can a model learn, from a set of training templates, a representation of text, layout and spatial relationships general enough that it extracts key-value fields correctly from a template it has never seen, with no labelled or unlabelled data from that template used at any point in training?

The question matters because the alternative scales badly, and because it is a clean test of what a document model has actually learned. A model that memorises "the invoice number is in the top right" will score well on a held-out *document* of a template it trained on and fail on a new layout. Only a template-level split can tell these apart, and most reported numbers in this area do not use one.

Our answer has three parts. First, we pose the task **schema-free**: the model is never given a list of field names, so it cannot overfit to a fixed vocabulary of fields and can therefore be asked about forms whose fields it has never heard of. Second, we make the representation **geometrically relative**: the component that decides which key a value answers sees only offsets between the two boxes, measured in units of the page's own text size, and we prove this is invariant to where the form sits on the page and how large it is scanned. Third, we measure **honestly**: splits are built over induced templates with leakage assertions that fail loudly, confidence intervals resample templates rather than documents, and metrics that a dataset cannot support are reported as not-measurable rather than as zero.

Structure of this report. Section 2 outlines the project workflow. Section 3 surveys the material we reviewed. Section 4 describes the proposed approach, split into the work done before the prep-presentation review (Section 4.1) and after it (Section 4.2). Section 5 gives the dataset details. Section 6 describes every experiment performed, including those with negative results. Section 7 presents the results. Section 8 gives the plan for novelty assessment, and Section 9 concludes.

Code and trained models. The complete source, configuration and this report are at `https://github.com/MudiamSukesh/cross-template-kv-extraction`. The four trained checkpoints are published as release assets at `https://github.com/MudiamSukesh/cross-template-kv-extraction/releases/tag/v1.0`.

2 Project Workflow

- **Problem framing.** Fix the task as cross-template key-value extraction with zero same-template training data; identify that a template-level split is the only valid evaluation.
- **First attempt (pre-review).** A schema-driven, query-based extractor: supply a field name as a query, have the model tag the span answering it. Abandoned; see Section 4.1.
- **Restart (post-review).** Re-pose the task schema-free as tag-and-link, with two heads on one backbone.
- **Data curation.** Four public corpora into one unified page-level schema; a validator that refuses malformed records; an alignment check that verifies boxes sit on printed ink.
- **Template induction and splitting.** Cluster documents into templates by layout signature, then split *templates* (not documents) into train, validation, seen test and unseen test, with leakage assertions.
- **Modelling.** LayoutLMv3 backbone; a BIO role tagger; a link scorer using relative geometry; joint training with two losses.
- **Evaluation.** Seen-versus-unseen gap with template-level bootstrap intervals; VRDU's own benchmark-style field scoring; per-field and per-source breakdowns.
- **Failure analysis.** Counterfactual layout probes and structural bucketing to name the assumptions that fail rather than merely reporting that they do.
- **Ablations.** One configuration key each, to test whether the generalisation techniques earn their place.
- **Interface.** A web application that accepts an uploaded image or PDF, runs OCR and the model, and draws the extracted pairs on the page.

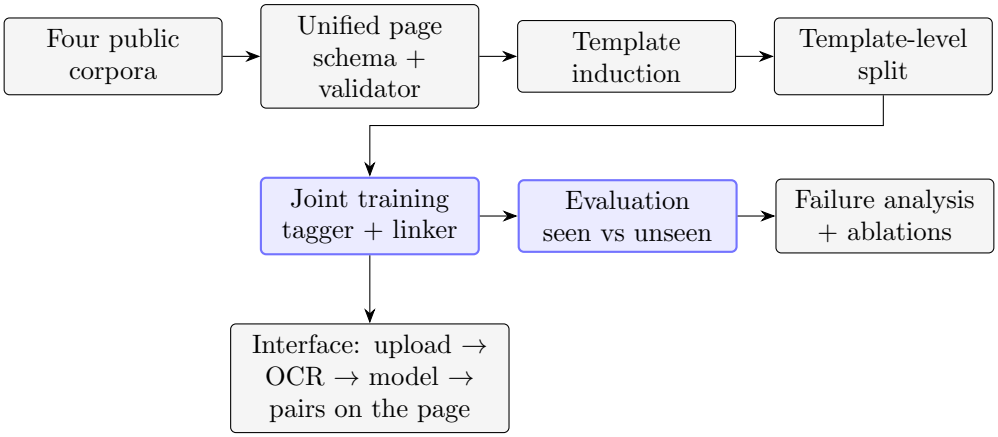


Figure 1: Project workflow. Shaded boxes are the two learned stages; everything else is data preparation or measurement. VRDU enters only at the evaluation stage and never at training.

3 Literature Survey

We restrict this survey to the material the project directly builds on.

Layout-aware pre-trained backbones. The LayoutLM family established that a transformer over document text can be made layout-aware by adding the token's bounding box as a positional input. LayoutLM [1] added 2D position embeddings to BERT. LayoutLMv2 [2] added a visual stream and a spatial-aware self-attention bias. LayoutLMv3 [3] simplified the recipe with unified text and image masking and word-patch alignment, and is the backbone we use. The property we rely on is that the model consumes a sequence of (token, box) pairs and produces contextual embeddings in which a word's neighbours on the *page* influence it, not only its neighbours in reading order. We note the detail that each box coordinate is normalised to $[0, 1000]$ against the page's own width and height independently, which distorts aspect ratio — a point that motivates our own geometry handling in Section 4.2.3.

Form-understanding datasets. FUNSD [4] is the reference dataset for the task we solve: 199 noisy scanned forms annotated with entities typed as question, answer or header, *and* with explicit links between them. It is the only corpus we use that supervises the linking decision directly. SROIE [5] provides 973 scanned receipts with four extracted field values per receipt but no key annotations and no links, which makes it value-only supervision. WildReceipt [6] contributes 1,740 receipt images with 25 key-information categories paired as `X_key`/`X_value`, from which links can be derived.

The cross-template benchmark. VRDU [7] is the work closest to our problem statement. It argues that existing benchmarks overstate generalisation because their train and test documents share templates, and it supplies two corpora — a registration-form corpus derived from US FARA filings and an ad-buy-form corpus derived from the DeepForm project — together with explicit few-shot splits in which the test documents come from templates *unseen* during training. It also defines a careful field-matching protocol in which the comparison between an extracted string and a gold one is type-specific: dates are compared after parsing, prices within a tolerance, addresses under an edit-distance threshold, and so on. We adopt both the corpora and this matching protocol, porting the published matching and scoring code so that our numbers are comparable in kind to those reported there. Crucially, VRDU annotates *values only*: it does not mark which printed text on the page is the key, nor link keys to values. This shapes what we can and cannot measure, as Section 7.1 explains.

OCR. Uploaded documents arrive as pixels, so the system needs its own text detection and recognition. We use docTR [8] with the FAST detector [9] and the PARSeq [10] recogniser.

PARSeq's permuted autoregressive formulation gives notably better recognition on the short, noisy strings typical of scanned forms than the CRNN baseline docTR falls back to, which is why the system verifies its weights are present and says so in the interface when they are not.

4 Proposed Approach

4.1 Work done before prep-presentation review

The first design was **schema-driven and query-based**. A field name — say `signer_name` — was embedded as a query prepended to the page's tokens, and the model tagged the span of words that answered that query, allowing multiple spans per query and up to 400 candidates. The backbone was LayoutLMv3-base, as it remains. A candidate-pruning stage reduced the span search space before scoring, and the tag head used focal loss to counter the severe class imbalance between answer words and the rest of the page.

This design was implemented, trained through several iterations and evaluated. It was then **abandoned**, for reasons that are themselves a result and are reported in Section 6.2. In brief: the tag head collapsed on unseen templates, scoring 0.019 on the FARA corpus, the pruning stage capped achievable recall at 0.822 regardless of what the model did downstream, and the per-query formulation could not resolve the common case of several instances of the same field type on one page — three different names on a single form, all plausible answers to a query for "name". A proposed fix was to anchor each query to the *bounding box* of the printed key rather than supplying the key as text alone, turning the question from "what is the value called name?" into "what is the value of the entity at this location?". That idea survives, in a different form, as the link head of the current design.

The deeper problem was structural. A query-based model must be told what to ask for, which re-introduces exactly the per-template, per-schema configuration cost that the project exists to remove. At the review it was clear the approach had become complicated without addressing the central question, and the decision was taken to restart.

4.2 Work done after prep-presentation review

The system was re-posed **schema-free**. The model is never told which fields exist. It answers two questions about a page, and the deliverable is whatever key-value pairs the page itself contains.

4.2.1 Formulation

Head 1 — role tagging. Every word is assigned one of seven BIO tags: `O`, plus `B-` and `I-` for each of KEY, VALUE and HEADER. A single linear layer maps the backbone's hidden state h_t of a word's first sub-token to tag logits,

$$z_t = W h_t + b, \qquad W \in \mathbb{R}^{7\times 768},$$

and spans are decoded by walking the sequence. Decoding is deliberately lenient: an `I-X` with no open span opens a new one rather than being discarded, because a stray `I-` without its `B-` is a common small error and recovering a span from it loses less than throwing the word away.

Head 2 — linking. Given the key spans and value spans on a page, the model decides which key each value answers. A span's embedding is the mean of its words' hidden states, $\bar{h}_k$ and $\bar{h}_v$,

projected to a smaller entity space. The score of pairing key k with value v is

$$s(k, v) \;=\; \mathrm{MLP}\big(\big[\, e_k \;;\; e_v \;;\; e_k \odot e_v \;;\; \varphi\big(g(k, v)\big) \,\big]\big)\,,$$

where e_k, e_v are the projected span embeddings, $e_k \odot e_v$ is their elementwise product (an interaction term that lets the model detect agreement between the two, which a concatenation alone cannot express), and $\varphi(g(k, v))$ is a learned embedding of the twelve relative geometry features of Section 4.2.3. A separate scalar

$$n(v) \;=\; w_n^{\top} e_v + b_n$$

scores the hypothesis that v answers *no* key on this page. For each value, the scores of all candidate keys together with $n(v)$ form a softmax:

$$p(k \mid v) \;=\; \frac{\exp\big(s(k, v)\big)}{\exp\big(n(v)\big) + \sum_{k' \in K} \exp\big(s(k', v)\big)}.$$

This is a per-value competition, which is the right shape for the data: one key may answer many values (a column header above a column), but a value answers at most one key.

4.2.2 The two losses

Training minimises

$$\mathcal{L} \;=\; \mathcal{L}_{\mathrm{tag}} \;+\; \lambda\, \mathcal{L}_{\mathrm{link}}, \qquad \lambda = 1.$$

The tagging loss is the mean cross-entropy over *supervised* words only,

$$\mathcal{L}_{\mathrm{tag}} \;=\; -\frac{1}{|S|} \sum_{t \in S} \log p\big(y_t = c_t \mid x\big),$$

where S excludes padding, non-first sub-tokens, words in `Ignore` segments, and — for value-only corpora such as SROIE and VRDU — every word that is not part of an annotated value, since on those pages we do not know whether any *other* word is a key.

The linking loss treats the gold key set $G(v)$ of a value as a set of acceptable answers and maximises their total probability:

$$\ell(v) \;=\; -\log \sum_{k \in G(v)} p(k \mid v),$$

computed with `logsumexp` for numerical stability. Values whose gold key falls in a different chunk are skipped rather than being scored as errors with an unreachable target.

4.2.3 Relative geometry, and why it is the load-bearing idea

LayoutLMv3 normalises each box coordinate to $[0, 1000]$ against the page's width and height *separately*. A square stamp on a portrait page therefore becomes a non-square box, and the same form scanned at a different size or offset produces different numbers. If the link head saw those coordinates, it could learn "the value is at $x \approx 700$", which is true of the training templates and false of everything else.

So the link head never sees absolute coordinates. It sees twelve features computed in an aspect-preserving frame (the longest page side mapped to 1000, the other scaled by the same factor) and expressed in units of the page's *median word height* m: horizontal and vertical centre offsets, the

gap between the boxes, their width and height ratios, horizontal and vertical overlap fractions, and indicators of reading-order direction. Each unbounded offset is compressed by a signed log,

$$\operatorname{slog}(x) \;=\; \operatorname{sign}(x)\,\log\big(1+|x|\big),$$

which keeps a one-word gap and a half-page gap on the same scale without discarding the sign.

Invariance. Let a page be transformed by a translation and a uniform scaling, $x' = \alpha x + \beta_x$ and $y' = \alpha y + \beta_y$. Any centre difference becomes

$$\Delta x' \;=\; (\alpha x_k + \beta_x) - (\alpha x_v + \beta_x) \;=\; \alpha\,\Delta x,$$

so the translation cancels. The median word height transforms as $m' = \alpha m$, so each feature, being a ratio of a length to m, satisfies

$$\frac{\Delta x'}{m'} \;=\; \frac{\alpha\,\Delta x}{\alpha\, m} \;=\; \frac{\Delta x}{m}.$$

Overlap fractions and aspect ratios are already dimensionless and likewise unchanged. The twelve features are therefore *identical* for a form photographed close up and the same form scanned small in a corner, which is exactly the nuisance variation that distinguishes one template's rendering from another's. Section 7.6 shows this is not merely a nice property: replacing it with absolute geometry improves seen-template accuracy and measurably destroys cross-template transfer.

4.2.4 Adaptation method

The adaptation technique is **full fine-tuning** — every one of the 126.0M backbone parameters is updated, alongside 0.70M parameters in the two heads — rather than LoRA, adapters or a frozen backbone. Full fine-tuning was chosen because the task changes what the *representation* must encode (spatial relationships between entities, not just token semantics), and a frozen backbone cannot adapt its attention to that. The risk of full fine-tuning on a small corpus is memorisation of the training templates, so it is constrained by five safeguards, listed in Table 1.

Safeguard	**Why**
Split learning rates	3×10^{-5} backbone, 3×10^{-4} heads: the pre-trained weights are good and should move slowly; the randomly initialised heads must move fast.
Warmup then linear decay	10% warmup stops large early updates from destroying pre-trained features before the heads produce a useful signal.
AdamW, weight decay 0.01	Excluding biases and LayerNorm parameters, which should not be shrunk toward zero.
Gradient clipping at 1.0	Caps the damage of any single bad batch.
Early stopping on validation *templates*	Patience 5 on a mean of the validation metrics; "best" means best on layouts not trained on, so the stopping criterion itself rewards generalisation.

Table 1: The five constraints placed on full fine-tuning.

Geometric augmentation is applied during training: a global scale in $[0.85, 1.0]$, a random shift, and per-word jitter of 0.15 word heights. The intent is to prevent the *tagger* — which, unlike the link head, does see LayoutLMv3's absolute coordinates — from memorising positions.

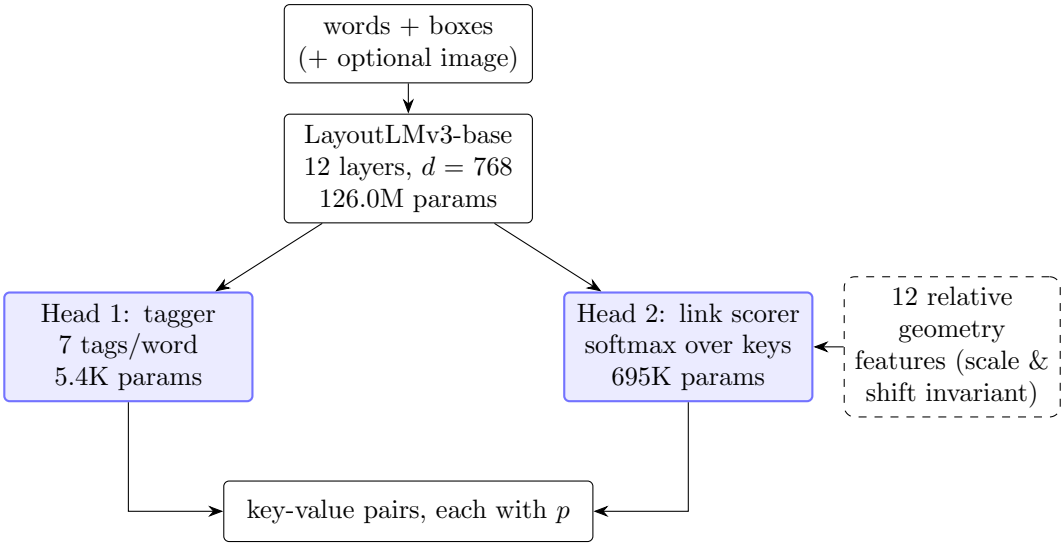


Figure 2: One backbone, two heads. Only the link head is geometry-invariant by construction; the tagger sees LayoutLMv3's absolute coordinates and is regularised by augmentation instead.

4.3 The interface

The deliverable includes a web application so that the method can be exercised on a document the model has never seen, rather than only through evaluation scripts. It is a Flask server started with `python app\app.py --run runs\base` and served at `127.0.0.1:5000`.

Methodology. The design constraint is that the interface must run the *same* code path as the evaluation, so that what a user sees cannot diverge from what is reported. It therefore imports the identical `kvx` modules: the same OCR wrapper, the same encoder, the same checkpoint, the same decoding. Nothing is reimplemented for display. The model and the OCR networks are loaded once at start-up and held resident; requests are served one at a time, because a single GPU cannot safely run concurrent inference.

An upload is handled as follows. The file is accepted only if it is an image, a PDF, or one of the annotation formats the loaders understand; anything else is rejected with HTTP 400 rather than being guessed at. A PDF is rendered to page images at 144 dpi with `pypdfium2`. Each page then goes through docTR, the chunker, the backbone and the two heads exactly as in Section 4.2, and the resulting spans and links are serialised into a per-page payload holding every word with its box, role and confidence; every predicted pair with its key box, value box and link probability p; and the OCR and model timings.

The header reports which checkpoint, backbone, device and OCR models are live, and says so explicitly when the PARSeq recogniser weights are absent and the cached CRNN fallback is in use — a silent fallback would otherwise make OCR quality look like model quality. Values that won no key are listed as `(no key found)` rather than being dropped, which keeps the display honest about what the model did and did not link.

5 Data set Details

Four public corpora are used. Three train; one is held out entirely and is never trained on, not even as unlabelled data.

Procurement. All four are public research datasets, obtained from:

View	What it does
Extract	Accepts a dropped image or PDF and draws the extracted pairs on the rendered page. Hovering a row highlights that key, that value and the link between them. A toggleable layer colours every word by predicted role with per-role counts, which exposes tagger behaviour rather than only the final pairs. Multi-page PDFs get page buttons.
Held-out examples	Serves pages the model was never trained on — VRDU documents from unseen templates, and seen-template test documents — with every gold value marked found or missed by the same matching function the metrics use, so the displayed verdict and the reported number cannot disagree.
Results	Renders `metrics.json` from the evaluation run directly: seen versus unseen with 95% intervals, the gap, per source and per field. It displays the measured file and computes nothing of its own.

Table 2: The three views of the interface.

Corpus	Role	Docs	Templates	Annotation
FUNSD	train	199	155	entities typed question/answer/header *and* links: full supervision
SROIE	train	973	307	four field values per receipt; no keys, no links
WildReceipt	train	1,739	1,530	25 categories as `X_key`/`X_value`; links derived
VRDU	**test only**	2,556	1,100	values only, with a type per field; no keys, no links

Table 3: The four corpora. VRDU contributes 4,991 annotated pages across its two corpora (registration-form, 1,915 documents and 8,826 gold items; ad-buy-form, 641 documents and 14,149 gold items).

- FUNSD [4] — https://guillaumejaume.github.io/FUNSD/
- SROIE [5], ICDAR 2019 Robust Reading Challenge Task 3 — https://rrc.cvc.uab.es/?ch=13
- WildReceipt [6], distributed with MMOCR — https://download.openmmlab.com/mmocr/data/wildreceipt.tar
- VRDU [7], Google Research — https://github.com/google-research-datasets/vrdu

None is redistributed with our code; the repository documents these sources and the build scripts consume whatever is placed in `data/`.

Unified schema. Every loader emits the same page-level record: a document id, a source tag, the page image path and size, a list of words with their boxes, a list of entities with a role and a span of word indices, and a list of links as (key entity, value entity) index pairs. Optional fields record whether the page is fully annotated or value-only, the OCR line grouping, words to ignore, and the gold field values used for field-level scoring. A validator refuses any record with out-of-range boxes, dangling entity indices or links pointing at non-existent entities, so a malformed corpus fails at build time rather than silently degrading training.

Pre-processing. Boxes are normalised to LayoutLMv3's $[0, 1000]$ convention. Values in value-only corpora are matched back onto the page's words by exact match first, then by a normalised fuzzy match, so that a gold value `9.00` is found inside the printed `RM9.00`. An alignment check rasterises a sample of pages and measures the share of boxes containing ink against a deliberately shifted control, and reports `ALIGNED` only when the real boxes contain substantially more ink than the control — this catches coordinate-order and origin errors that would otherwise look like model failure.

Template induction. None of the corpora ship template labels, so templates are induced. Each page is reduced to a layout signature — a coarse spatial histogram of its text boxes together with the set of frequent short strings printed on it — and pages are clustered by average-linkage agglomerative clustering under a distance combining layout and text-set dissimilarity. The threshold was fixed by inspecting FUNSD montages until clusters corresponded to visually identical forms. The resulting counts appear in Table 3.

The split. Splitting is over *templates*, not documents: a template's entire set of documents goes to exactly one of train, validation, seen test and unseen test. Seen test holds unseen *documents* of training templates; unseen test holds documents of templates no training document shares. After the split, assertions verify that no template id appears in more than one of train, validation and unseen test, and a near-duplicate check verifies that no held-out page lies within a small distance of any training page. These assertions run again inside the training entry point, so a leak cannot survive to a reported number.

6 Experiments

Hardware and software. A single NVIDIA GeForce RTX 5060 Laptop GPU under Windows 11, with PyTorch 2.11 built against CUDA 12.8, bfloat16 automatic mixed precision, and Python 3.13. One run of the base configuration takes roughly two hours; a full evaluation over all three splits takes about twenty minutes, and the VRDU benchmark scoring runs from saved predictions in about a minute without loading the model.

Training configuration. LayoutLMv3-base backbone (126.0M parameters) plus 0.70M head parameters; maximum 512 tokens per chunk with chunk boundaries placed only between units, so no key or value is ever split; batch size 2 with gradient accumulation 2; AdamW with

backbone learning rate 3×10^{-5}, head learning rate 3×10^{-4} and weight decay 0.01 excluding biases and LayerNorm; 10% warmup then linear decay; gradient clipping at 1.0; up to 20 epochs with early stopping at patience 5 on the mean of the validation metrics. Sources are sampled equally per epoch, so that WildReceipt's 1,231 training documents do not drown FUNSD's 126. The base run selected epoch 16.

6.1 Every experiment performed

Table 4 lists every configuration that was trained or scored during the project, including the approach that was abandoned and the four ablations that remain unrun. Each entry is one line of configuration away from the base run, which is what makes the comparisons in Section 7.6 interpretable.

Experiment	Status	Purpose / outcome
Query-based extractor	**negative; abandoned**	Section 6.2
`runs\base`	complete	the reported system
`runs\unseen15`	complete	holds out in-domain templates so that entity, link and end-to-end F1 become measurable on unseen templates
`runs\absgeom`	complete	absolute instead of relative link geometry
`runs\noaug`	complete	augmentation disabled
VRDU through our own OCR	complete	tests the gold-OCR assumption
VRDU benchmark scoring	complete	re-scores saved predictions under VRDU's own protocol
`nogeom`, `nobalance`, `nosroie`, `segment`	not run	listed in Section 8

Table 4: All experiments, including those that failed.

6.2 Negative results

The query-based extractor failed and was abandoned. Three measurements ended it. The tag head *collapsed on unseen templates*, scoring 0.019 on the FARA corpus while performing acceptably on templates it had trained on — the clearest possible symptom of template memorisation. The candidate-pruning stage had a recall ceiling of 0.822, meaning that almost one value in five was discarded before the model ever scored it, so no downstream improvement could have recovered them. And the per-query formulation had no mechanism to disambiguate several instances of one field type on a page. Focal loss on the tag head and a bounding-box anchor for the query were implemented as remedies; the restart was taken before they could be evaluated head to head, so we report them as unresolved rather than as fixes that did or did not work. The lasting lesson is the one in Section 4.1: a query-based model must be told what to ask for, which re-imposes the per-schema cost the project exists to remove.

Augmentation is not measurably useful. Disabling it changes nothing beyond noise (Section 7.6). It is retained on argument, not evidence, and this is stated rather than hidden.

VRDU's ad-buy line items are effectively not extracted, at 0.005 F1. This is a real failure of the approach on tables and is analysed in Section 7.8.

A field-naming bug was found by inspecting test-set output. Early VRDU scoring chose a field by the longest matching alias, so a key reading "Date of signature" matched `signer_name` (via "signature", nine characters) in preference to `file_date` (via "date", four), giving `file_date`

a recall of 0.144 against a value recall of 0.750. The fix ranks candidate fields by whether the *value*'s data type fits the field's matching function before falling back to alias length. We record explicitly that this fix was motivated by looking at test-set output, which is a form of test-set contact; any further mapping changes belong on VRDU's own validation lists.

7 Results

7.1 Metrics, and why some cells are empty

- **value_recall** — share of gold values the model produced as a value span anywhere on the page. Measurable on all four corpora, so it is the headline number.
- **kv_field_recall** — value found *and* linked to a key whose text maps to the right field name. Needs a field vocabulary: SROIE and VRDU only.
- **entity_f1**, **link_f1**, **e2e_pair_f1** — span quality, linking quality given gold spans, and the two composed. All three need annotated keys and links: FUNSD and WildReceipt only.

VRDU and SROIE annotate values only. A metric those corpora cannot support is printed as "–", never as zero, because zero would be a claim about the model and "–" is a statement about the data. All intervals are 95% bootstrap intervals that resample *templates*, since documents of one template are not independent observations.

7.2 Seen versus unseen templates

Split	Docs	Tmpl	value_rec	kv_field_rec	entity_f1	link_f1	e2e_pair_f1
val	310	197	0.946	0.728	0.933	0.940	0.814
seen_test	280	197	0.918	0.648	0.914	0.935	0.769
unseen_test (VRDU)	4,991	1,100	0.695	0.511	–	–	–

Table 5: The base run. Unseen test is VRDU in its entirety.

Metric	Seen	Unseen	Gap	95% CI	Relative drop
value_recall	0.918	0.695	+0.223	[+0.188, +0.253]	24.3%
kv_field_recall	0.648	0.511	+0.137	[+0.012, +0.273]	21.1%

Table 6: The generalisation gap of the base run.

7.3 Separating template novelty from domain novelty

Table 5 cannot fill its last three columns because every held-out template comes from VRDU. Re-splitting with 15% of FUNSD and WildReceipt templates held out and retraining (`runs\unseen15`) fills them, and the result is the most important in this report.

Only the two recall metrics fall, and the per-source rows locate the fall:

FUNSD is identical to three decimal places; WildReceipt is marginally better on unseen templates; SROIE loses 3.3 points. The aggregate gap of 0.189 is an artefact of composition, because 4,991 of the 5,430 unseen documents are VRDU, a domain with no training representation at all. **Template novelty is close to free; domain novelty is what costs.** The error breakdown

Metric	Seen	Unseen	Gap	95% CI	Relative
value_recall	0.918	0.728	+0.189	[+0.153, +0.220]	20.6%
kv_field_recall	0.711	0.479	+0.233	[+0.114, +0.354]	32.7%
entity_f1	0.897	0.913	−0.016	[−0.054, +0.020]	−1.8%
link_f1	0.949	0.942	+0.007	[−0.023, +0.037]	0.7%
e2e_pair_f1	0.788	0.778	+0.010	[−0.068, +0.097]	1.2%

Table 7: The `unseen15` run. **All three structural metrics have gap intervals containing zero**: they do not degrade on unseen templates.

Source	Seen value recall	Unseen value recall
FUNSD	0.894 (20 docs)	0.894 (32 docs)
WildReceipt	0.971 (69 docs)	0.975 (261 docs)
SROIE	0.813 (142 docs)	0.780 (146 docs)
VRDU-FARA	–	0.681 (3,632 docs)
VRDU-DeepForm	–	0.689 (1,359 docs)

Table 8: Within a source, unseen templates cost almost nothing.

agrees: 94.6% of gold-linked values take the right key on seen templates and 94.4% on unseen ones.

7.4 VRDU benchmark-style field scores

Re-scoring the saved predictions under VRDU's own type-specific matching and its precision/recall/F1 definition, with no VRDU training data of any kind, gives **0.593 micro F1** on registration-form (micro P 0.656, R 0.541, CI [0.567, 0.616]) and **0.497 unrepeated F1** on ad-buy-form. Per field, `registration_num` reaches 0.920 F1 and `registrant_name` 0.789, while `signer_title` scores 0.002 and `line_item` 0.005.

The `unseen15` run of Section 7.3 scores lower on the same metric — 0.516 micro F1 [0.437, 0.542] on registration-form and 0.456 unrepeated F1 on ad-buy-form — which is the expected cost of the 377 training documents it gives up in order to make the structural metrics measurable. The two runs answer different questions and neither supersedes the other: the base run is the system, and `unseen15` is the instrument that separates template novelty from domain novelty.

7.5 Per-field behaviour: finding is not keying

Three bands are visible in Figure 3. Fields with one stable printed label — `registration_num` (93/92), `contract_num` (84/83), `file_date` (75/74), `advertiser` (68/66) — have bars within three points of each other: once found, they are named correctly. Table columns collapse: `channel` falls from 0.777 to 0.218, `sub_amount` from 0.955 to 0.558, `gross_amount` from 0.977 to 0.690. Values with no printed label at all are never found in the first place: `signer_title` 0.078, `tv_address` 0.138, `signer_name` 0.246.

7.6 Ablations

`absgeom` **moves every measure in the predicted direction.** Absolute geometry is *better* than relative geometry on seen templates — 0.938 against 0.918 value recall — and worse on every measure that involves a template the model has not met: unseen value recall falls from 0.695 to 0.671, the seen-minus-unseen gap widens from +0.223 to +0.267, the FARA

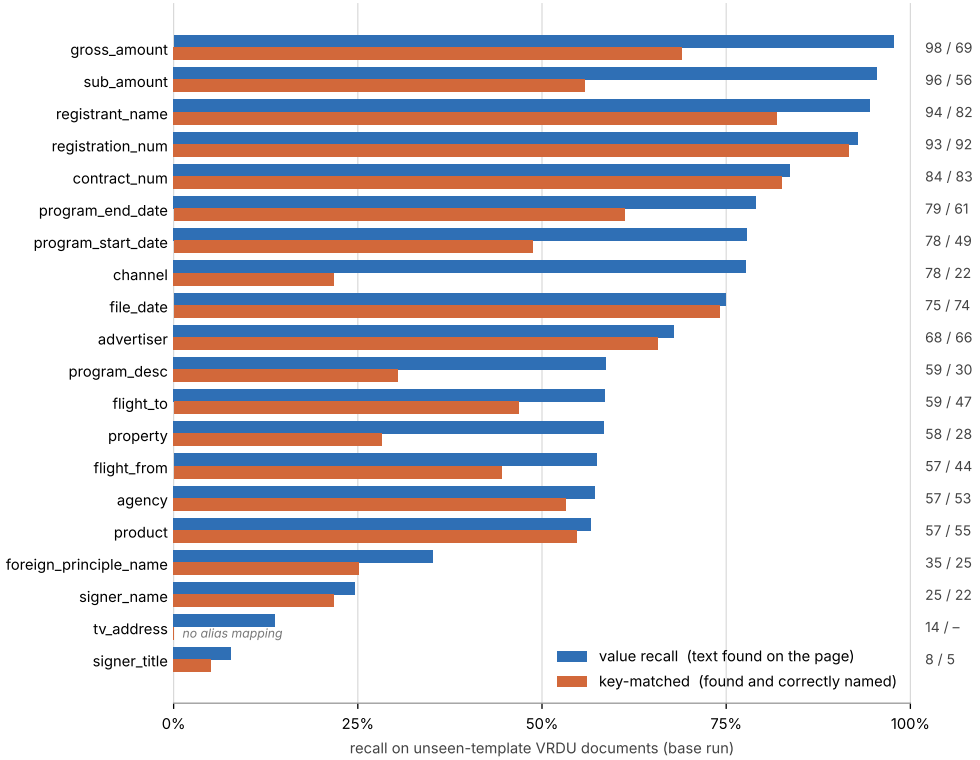


Figure 3: VRDU per-field recall on unseen templates, base run, both corpora pooled. The upper bar is value recall (the text was found); the lower bar is key-matched recall (found *and* correctly named). The distance between them is the share of values the model located but could not name. `tv_address` has no alias mapping and so has no key-matched bar.

Run	Seen	Unseen	Gap	FARA F1 [95% CI]	DeepForm F1
base (relative geometry + aug.)	0.918	0.695	+0.223	**0.593** [0.567, 0.616]	**0.497**
`absgeom` (absolute geometry)	**0.938**	0.671	+0.267	0.541 [0.452, 0.593]	0.484
`noaug` (no augmentation)	0.933	**0.703**	+0.230	0.571 [0.524, 0.608]	0.478

Table 9: Ablations. Seen and unseen columns are value recall.

field score drops from 0.593 to 0.541, and the DeepForm score from 0.497 to 0.484. Five independent measures, one direction. The natural reading is that absolute coordinates buy in-domain accuracy by learning where fields sit on the training templates, which is precisely the shortcut the relative frame is designed to remove; an ablation that merely removed capacity would have cost accuracy on both sides rather than trading one for the other.

No single one of those differences is statistically significant, and the report does not claim otherwise. The FARA intervals [0.567, 0.616] and [0.452, 0.593] overlap, as do the two gap intervals [+0.188, +0.253] and [+0.238, +0.298]. With one run per configuration and bootstrap intervals taken over templates, this experiment can establish a consistent direction but not significance; distinguishing the two would need several seeds per configuration, which was beyond the compute available. The honest summary is that the geometry result is *directionally consistent across five measures and two corpora, and individually inconclusive on each.*

`noaug` **is weaker still.** Its FARA interval [0.524, 0.608] sits almost entirely inside the base interval, and unseen value recall is in fact marginally *higher* without augmentation (0.703 against 0.695). Scale, shift and jitter make no measurable difference on this test set. They cost nothing and are kept on the argument given in Section 4.2 rather than on evidence, and that is a weaker

footing than the geometry claim.

7.7 Qualitative results

Numbers alone do not show whether a prediction is sensible, so three artefacts are produced alongside them. Step 3 renders gold annotations onto sample pages, which is how the template clustering threshold was fixed by eye. Step 5 writes a `qualitative/` directory per run holding predicted pages with key and value boxes drawn and each link joined, sampled from both the seen and the unseen splits. Step 6 writes an `errors/` directory containing the individual failure cases behind each error category in the tables above, so that a claim such as "wrong key, nearer than gold" can be checked against the page that produced it.

The interface of Section 4.3 is the interactive form of the same thing: its held-out examples view marks every gold value found or missed on documents from unseen templates, using the same matching function as the metrics, and its extract view shows the model's output on a document supplied by the user.

7.8 What limits generalisation

Two structural assumptions fail, and the failure analysis names both.

"A key sits beside its value" fails for tables. A column header is one key for dozens of values stacked beneath it, which is not the one-key-beside-one-value geometry the link scorer was trained on. This produces the found-but-not-keyed collapse of Figure 3 and the 0.005 line-item F1. Counterfactual probes confirm the direction bias directly: moving every value below its key costs 0.167 of value accuracy, and mirroring the page horizontally costs 0.227.

"Every value has a printed label" fails for signature and address blocks. Here the first bar is already low — the tagger does not mark the span at all — because nothing on the page says what the text is. A schema-free model cannot invent a field name, so this is an honest ceiling of the approach rather than a tuning gap. Masking all key text drives end-to-end pair F1 to 0.000, confirming how much of the decision rests on what keys say.

A third, softer limit is multi-line and long values: recall falls from 0.739 on single-line values to 0.265 on multi-line ones, and from 0.819 on one-word values to 0.398 on values of five words or more.

8 Plan for Novelty Assessment

The failure analysis points at three changes that are principled rather than tuning, in that each targets a named structural assumption and uses machinery the model already has.

1. **Group table cells into rows by vertical banding rather than reading order.** Targets the 0.005 line-item F1. Reading-order grouping breaks whenever one cell is missing or extra; banding by shared vertical position is how a person reads a table row.
2. **Allow a key above a column of values**, weighting the column-overlap feature accordingly. Targets the found-but-not-keyed collapse. The geometry features already carry horizontal overlap; the training data simply contains very few table examples, so the relation is never learned.
3. **Permit a value whose nearest label sits *below* it**, as in signature blocks. Targets `signer_title` at 0.078. The direction features exist and nothing in the model forbids this; it

was never taught.

All three must be developed against the in-domain unseen split or VRDU's own validation lists, never against the VRDU test documents reported here. Alongside them, four ablations remain unrun — `nogeom` (is linking carried by geometry or by key text?), `nobalance` (does equal source sampling matter?), `nosroie` (does value-only supervision help at all?) and `segment` (how optimistic is the gold-segment box protocol against honest per-word boxes?).

9 Conclusion

We built a schema-free key-value extraction system that is told nothing about which fields a form contains, and evaluated it under a template-level split with explicit leakage assertions. On templates it has never seen it recovers 0.695 of gold values against 0.918 on seen templates, and scores 0.593 micro F1 under VRDU's own benchmark-style field metric with no VRDU training data of any kind.

Three conclusions are supported by the measurements rather than by argument.

Relative geometry appears to be what makes the representation transfer. The `absgeom` ablation scores *better* than the full system on seen templates and worse on every cross-template measure — unseen value recall, the width of the gap, and both VRDU corpora. That asymmetry is the signature of a removed shortcut rather than of reduced capacity. The intervals overlap, so the result is directionally consistent across five measures rather than individually significant; confirming it would need several seeds per configuration.

Template novelty is nearly free; domain novelty is the real cost. Under a split that holds out in-domain templates, entity F1, link F1 and end-to-end pair F1 show no degradation at all — every gap interval contains zero — while value recall falls only on the cross-domain VRDU documents. The headline gap of 0.223 is therefore a statement about domain shift, and reading it as a template-generalisation gap would be wrong.

The failure modes are structural and nameable. The model assumes a key sits beside its value and that every value carries a printed label. Tables break the first; signature and address blocks break the second. Both were identified by counterfactual probes rather than by inspection.

Future work follows directly: the three changes in Section 8, the four remaining ablations, and an evaluation on forms from a third domain to test whether the domain-shift conclusion holds beyond VRDU.

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